

Medical QA RAG with Faithfulness Evaluation: Retrieval-Augmented Generation for Biomedical Question Answering with Domain-Specific Faithfulness Assessment

Federica Pasquali
OPIT - Open Institute of Technology
M.Sc Responsible AI

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Abstract

For the assessment I deployed an end-to-end Retrieval-Augmented Generation (RAG) pipeline for biomedical question answering, evaluated on PubMedQA [1]. The system compares three retrieval strategies, BM25 [2] (sparse), MedCPT [3] (dense), and a hybrid via Reciprocal Rank Fusion [4], followed by LLM-based answer generation. Result evaluation employs standard QA accuracy as well as a faithfulness evaluation that uses a domain-specific biomedical NLI model. Key finding is that the use of accuracy alone is insufficient as only 45.6% of answers from the best pipeline are both correct and faithful to the retrieved evidence, with 37% of correct answers relying on ungrounded reasoning. This result is a significant concern for clinical deployment.

1 Introduction and Problem Definition

The project centres on addressing biomedical question answering using a Retrieval-Augmented Generation (RAG) pipeline. Given a clinical research question, the system retrieves the relevant evidence from a corpus of PubMed abstracts, and it generates a grounded answer. Evaluation is conducted in terms of retrieval quality, answer accuracy, and faithfulness to the retrieved evidence.

Medical QA is a high-stakes NLP application where factual errors carry real consequences, because unsupported answers may can influence treatment decisions or create false confidence in unsupported conclusions. That is why faithfulness evaluation is used as it becomes a patient safety requirement for this application. The RAG architecture is choosen over alternatives based on this requirement, as fine-tuned classifiers could predict labels but offer no transparency into its reasoning, and a pure LLM could generate fluent answers with no guarantee of evidentiary grounding. RAG, by retrieving passages first and conditioning generation on them, it creates an auditable chain from question-evidence-answer that can be evaluated for faithfulness [5]. This design choice is supported by recent literature with MedRAG [6] that demonstrated that RAG can improve LLM accuracy by up to 18% across medical QA benchmarks.

1.1 Dataset: PubMedQA

PubMedQA [1] is a biomedical QA dataset collected from PubMed abstracts. The task is to answer research questions with yes/no/maybe using the corresponding abstracts. The dataset contains:

Table 1: PubMedQA dataset subsets and their roles.

Subset	Size	Labels	Role in this project
PQA-L (Labeled)	1,000	Expert-annotated (MD candidates)	Evaluation (500 dev / 500 test)
PQA-U (Unlabeled)	61,249	None	Retrieval corpus

Each instance contains a research question (from the article title), structured abstract sections (context), the abstract’s conclusion (long answer), and, for PQA-L only a yes/no/maybe label annotated by MD candidates. PQA-L contexts are added to the retrieval corpus to ensure a gold-standard passage exists for each evaluation query, with zero PubID overlap between subsets, as verified during data quality checks.

The dataset has a few limitations. It is English-only, restricted to PubMed-indexed research, and the yes/no/maybe framing simplifies real clinical reasoning. The class distribution is imbalanced with the following proportions ‘yes’ at 55.2%, ‘no’ at 33.8%, and maybe at ‘11.0%’, with a 5:1 majority-to-minority ratio. The original paper reports human accuracy of 78% and best-model accuracy of 68.1%, providing a performance context for this assessment results.

2 Exploratory Data Analysis

Comprehensive EDA was performed on both the evaluation set PQA-L and the retrieval corpus PQA-U to understand data characteristics and subsequent pipeline design decisions. Seven visualisations were produced; key findings are summarised below.

Label distribution: The majority class (‘yes’) accounts for 55.2% of instances, establishing a majority-baseline accuracy of 55.2%, so the models must exceed this to have value. This imbalance motivates the use of macro F1 as the primary evaluation metric to weight all classes equally.

Table 2: Summary statistics for PQA-L and PQA-U.

Statistic	PQA-L (n=1,000)	PQA-U (n=61,249)
Mean question length	12.9 words	13.3 words
Mean context length	200.2 words	198.7 words
Mean long answer length	39.7 words	42.0 words
Median sections per abstract	3	3
Unique MeSH terms	3,408	-
Mean MeSH terms per article	14.5	-

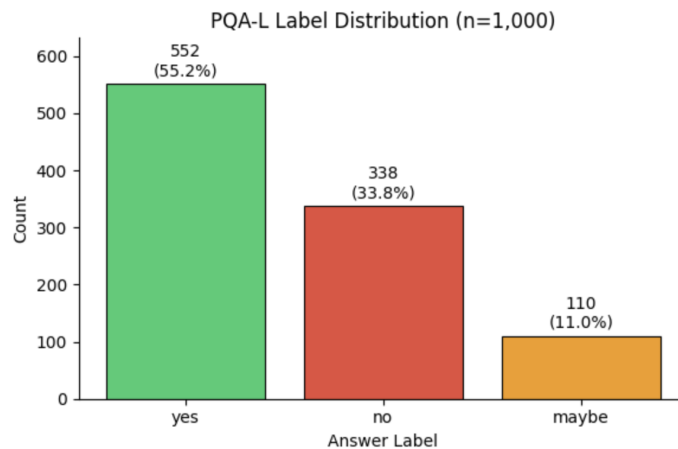


Figure 1: PQA-L label distribution (n=1,000) showing the 5:1 class imbalance between the majority class (‘yes’, 55.2%) and minority class (‘maybe’, 11.0%).

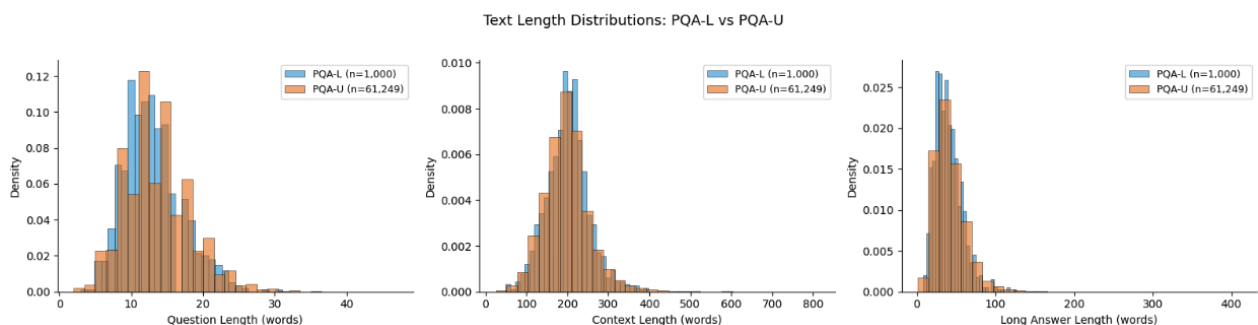


Figure 2: Text length distributions for PQA-L and PQA-U across questions, contexts, and long answers, confirming near-identical distributions between the evaluation and retrieval subsets.

Text characteristics: Questions are concise, averaging around 13 words, predominantly using a yes/no structure. 'Does', 'Is', 'Can' account for more than 50% of opening words, aligning with the length of research titles, with only a few questions starting with non-interrogative words ("Preoperative", "Treatment", "Delayed"), probably because some titles are statements phrased as implicit questions. The answer type is determined by the content of the evidence, not the complexity of the question. The retriever must find the right passage, not infer the answer only from the question structure.

The vast majority of abstracts contain three sections (mode=3 in both subsets: BACKGROUND, METHODS, RESULTS). Contexts average 200 words across 3 sections, short enough to treat each abstract as a single retrieval passage without chunking as section-level chunking would produce fragments too short for meaningful semantic matching.

Both subsets show near-identical distributions, confirming PQA-U is a suitable retrieval corpus for PQA-L queries.



Figure 3: Vocabulary comparison between questions (left) and contexts (right).

Vocabulary analysis: Questions use action-oriented language ('predict', 'affect', 'improve', 'reduce') while contexts use empirical language ('compared', 'group', 'significantly') reflecting how studies report findings. PQA-U covers 83.8% of PQA-L's vocabulary with a 16.2% gap that includes rare medical terms for which dense retrieval's semantic matching is valuable. The MeSH term analysis revealed broad coverage across clinical domains (3,408 unique terms), from retrospective studies to treatment outcomes and risk factor analysis, meaning that the retriever will have to handle diverse clinical domains.

The main key takeaways from the EDA for the pipeline design include:

- The use of macro F1 as primary metric due to the class imbalance.
- Adoption of single passage retrieval with no chunking because of short context (200 words).
- The partial vocabulary overlap justify the use of both lexical (BM25) and semantic (MedCPT) retrieval.

- The MeSH term analysis highlighted highly diverse medical domains, requiring domain-specific models over general-purpose ones.

3 Methodology

3.1 Data Parsing

Each PubMedQA instance contains a structured context field with variable abstract sections and their labels. These have been parsed into flat records by concatenating all context sections into a single passage string per abstract, preserving section labels and MeSH terms as metadata for analysis. Before data cleaning, quality checks confirmed no missing values, duplicate PubIDs, empty contexts, or PubID overlap between PQA-L and PQA-U.

The retrieval corpus was constructed by combining PQA-U (61,249 passages) with PQA-L (1,000 passages), generating 62,249 total passages. PQA-L was split into dev (500) and test (500) sets via stratified random split to preserve the 55.2/33.8/11.0% label distribution in both subsets.

3.2 Text Cleaning

Text preprocessing was minimal and deliberate, informed by the inspection of raw data quality. No HTML tags were found (apparent matches were mathematical comparison operators like $p < 0.001$), 21% of contexts contained non-ASCII characters (medical symbols: $\pm$, $\geq$, β , μ), and $< 0.2\%$ had double spaces. Cleaning consisted of: (1) normalising Unicode whitespace variants to standard spaces; (2) collapsing multiple spaces; (3) stripping leading/trailing whitespace. Medical symbols, comparison operators, and accented characters were kept. Additional preprocessing was applied at the component level because each pipeline component has different requirements. BM25 uses lowercased, stopword-removed tokens without stemming, to preserve terms like 'fibrillation'. MedCPT and the NLI model use their internal WordPiece tokenisers, while the LLM generator receives original-cased text. This approach avoids information loss from preprocessing and ensures that each component receives appropriately formatted input.

3.3 Chunking Strategy

Each abstract is treated as a single retrieval passage. This is supported by the EDA findings:

1. Abstracts averaging only 200 words, within the input limits of both BM25 and MedCPT (512 tokens).
2. The 264 unique label combinations make section-level segmentation inconsistent.
3. Section-level chunking would produce fragments of 60 words, too short for meaningful semantic matching.

This approach is consistent with the MedRAG benchmark [6], which indexes PubMed as one snippet per abstract.

3.4 Pipeline Architecture

Table 3: Pipeline components with domain specificity.

Component	Implementation	Domain-Specific
Sparse Retrieval	BM25 (rank_bm25)	N/A
Dense Retrieval	MedCPT + FAISS	Yes - trained on PubMed search logs
Hybrid Retrieval	BM25 + MedCPT via RRF (k=60)	Yes
Generation	Claude Sonnet API (temp=0)	General-purpose
Faithfulness	PubMedBERT-MNLI-MedNLI	Yes - biomedical NLI

Retrieval: BM25 [2] ranks passages by term frequency and inverse document frequency. It excels at exact keyword matching, which is advantageous in biomedical text where clinical terminology is precise and distinctive. MedCPT [3] is a biomedical retriever trained on 255 million PubMed user search logs using contrastive learning, with separate query and document encoders. The query encoder understands how researchers phrase questions, while the document encoder identifies the semantic content of abstracts.

The corpus, consisting of 62,249 passages, was encoded into 768-dimensional vectors and indexed with FAISS, IndexFlatIP with L2 normalisation for cosine similarity. The hybrid strategy combines both via Reciprocal Rank Fusion [4], summing inverse ranks with $k=60$. For each query, the top-5 passages are retrieved.

Generation: Retrieved passages are fed to Claude Sonnet with a structured prompt. The system instruction constrains the model to reason only from the provided evidence and produce a yes/no/maybe classification with 2-3 sentences of reasoning and answer 'maybe' when evidence is insufficient. Temperature is set to 0 for deterministic, reproducible outputs.

Faithfulness evaluation: Generated reasoning is assessed for scientific grounding using PubMedBERT fine-tuned on MNLI and MedNLI [7]. The top-3 retrieved passages serve as the NLI premise, truncated to 512 tokens, while the generated reasoning acts as the hypothesis. Each answer is classified as entailed (faithful), contradicted, or neutral.

4 Results

All results below are reported on the held-out test set (n=500). Design decisions, including prompt tuning and top-k selection, were made using the dev set (n=500). Both sets were created via stratified splitting to preserve label proportions.

4.1 Retrieval Performance

Table 4: Retrieval performance on the test set (n=500).

Retriever	Recall@1	Recall@3	Recall@5	Recall@10	MRR
BM25	0.686	0.812	0.846	0.886	0.755
MedCPT	0.656	0.836	0.884	0.924	0.754
Hybrid	0.764	0.878	0.924	0.968	0.834

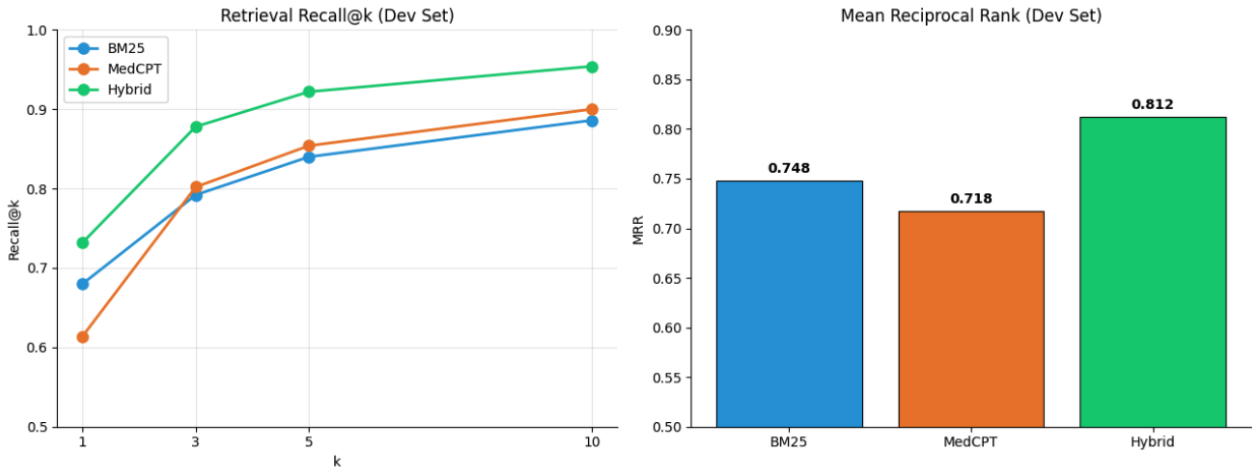


Figure 4: Retrieval Recall@k curves (left) and Mean Reciprocal Rank comparison (right) across BM25, MedCPT, and hybrid retrievers on the dev set (n=500).

BM25 achieves higher Recall@1 (0.686 vs 0.656) than MedCPT, indicating that exact lexical matching is effective when queries and passages share clinical terminology. But MedCPT surpasses BM25 at deeper ranks (Recall@10: 0.924 vs 0.886), capturing semantically relevant passages lacking keyword overlap. The hybrid retriever consistently outperforms both (Recall@5: 0.924, MRR: 0.834), confirming the complementarity of lexical and semantic signals and aligning with MedRAG findings [6].

4.2 QA Performance

Table 5: QA performance on the test set. Macro F1 is the primary metric due to class imbalance (5:1 ratio).

Method	Macro F1	Accuracy	Macro Prec.	Macro Rec.
Majority Baseline	0.237	0.552	0.184	0.333
BM25 + LLM	0.589	0.672	0.606	0.589
MedCPT + LLM	0.558	0.652	0.571	0.563
Hybrid + LLM	0.577	0.684	0.607	0.574

Table 6: Per-class metrics for the hybrid retriever on the test set.

Class	Precision	Recall	F1-Score	Support
yes	0.83	0.80	0.81	276
no	0.85	0.63	0.72	169
maybe	0.14	0.29	0.19	55

All RAG pipelines substantially outperform the majority baseline (macro F1: 0.237). BM25 achieves the highest macro F1 (0.589), followed by hybrid (0.577) and MedCPT (0.558). While the hybrid retriever leads on accuracy (68.4%) (comparable to the original PubMedQA best result of 68.1% [1] (BioBERT with multi-phase fine-tuning, without any fine-tuning on our part)) macro F1 better reflects performance across all classes given the 5:1 imbalance. Per-class analysis shows a difficulty gradient yes (F1=0.81), no (F1=0.72), maybe (F1=0.19). The 'maybe' class is the dominant failure mode, with the model over-predicting it when the context is noisy.

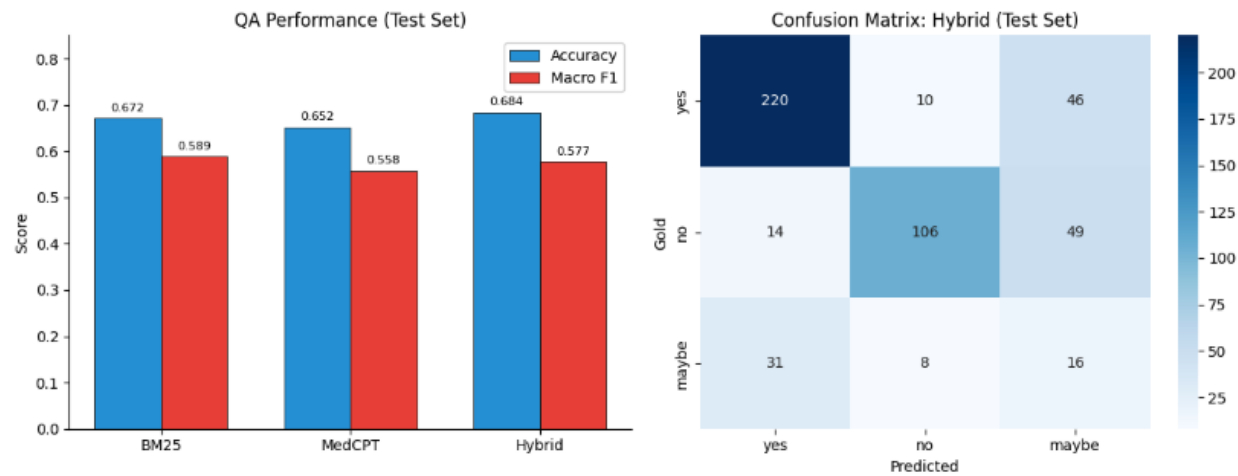


Figure 5: QA performance comparison across retrievers (left) and confusion matrix for the hybrid retriever (right).

4.3 Faithfulness Evaluation

Table 7: Faithfulness evaluation on the test set.

Retriever	Entailment	Neutral	Contradiction	Correct & Faithful
BM25	56.6%	3.2%	40.2%	39.8%
MedCPT	60.8%	4.4%	34.8%	40.4%
Hybrid	63.2%	2.6%	34.2%	45.6%

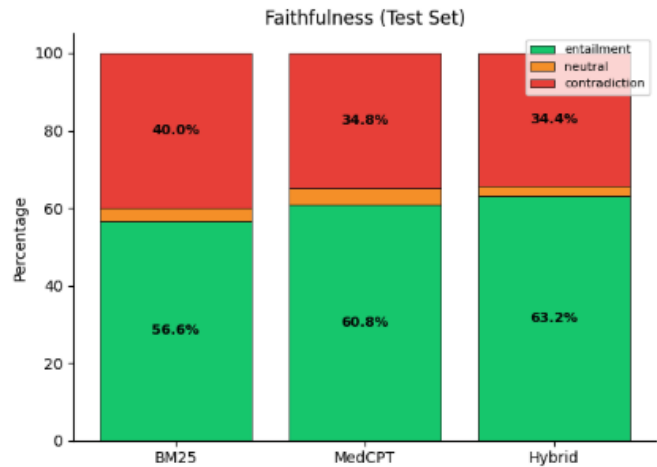


Figure 6: Faithfulness analysis: distribution by retriever (left), correctness vs faithfulness cross-analysis for the hybrid retriever (centre), and entailment score distributions for correct vs incorrect answers (right).

Only 45.6% of the hybrid pipeline’s answers are both correct and faithful. The correctness-faithfulness cross-analysis on the dev set revealed that 37% of correct answers were unfaithful meaning that the model arrived at the right answer for reasons not supported in the retrieved evidence. This ‘correct but unfaithful’ pattern is a reliability concern: a system that produces correct answers for the wrong reasons will fail unpredictably on novel queries.

4.4 Error Analysis

Impact of retrieval quality: When the gold passage was retrieved, accuracy reached 72.4% and faithfulness 62.0% but when it was missed, these dropped to 18.9% and 32.4%, respectively, a clear indicator that retrieval quality is a crucial factor affecting performance.

Error rate scales inversely with label certainty. ‘Faithful but incorrect’ cases primarily involve predicting ‘yes’ when the gold label is ‘maybe’, the model finds supporting evidence and commits to an affirmative answer when human annotators judged the evidence as inconclusive. ‘Maybe’ false positives (87 on dev) are split equally across ‘yes’ (46) and ‘no’ (41) gold labels, which suggests teh model uses ‘maybe’ as a general qualifier and not as a genuine uncertainty signal.

5 Dscussion

Retrieval as the primary lever. The accuracy difference between gold-passages-retrieved (72.4%) and gold-passages-missed (18.9%) queries confirms that retrieval quality plays a crucial role in downstream performance. This indicates that investing in better retrieval instead of larger generators is the most effective path to improving medical RAG systems. The hybrid retriever’s consistent advantage across all metrics validates the RRF fusion approach,

Table 8: Error rates and faithfulness by gold label (hybrid, dev set).

Gold Label	Error Rate	Faithfulness Rate
yes	20.7%	63.4%
no	37.9%	54.4%
maybe	67.3%	58.2%

consistent with findings from Self-MedRAG and the MedRAG benchmark [6].

The faithfulness gap. That only 45.6% of answers are both correct and faithful raises fundamental questions about RAG evaluation methodology. Standard QA benchmarks, including PubMedQA’s own leaderboard, evaluate only accuracy, missing the dimension of whether the model actually uses the provided evidence. The obtained results here suggest that published RAG accuracy numbers may overstate the system’s true reliability, as a considerable fraction of ‘correct’ answers may be ungrounded.

The ‘maybe’ problem. With an F1 of 0.19, the ‘maybe’ class is the dominant challenge. Determining when evidence is sufficient for a definitive answer versus only suggestive is still an open challenge, which requires calibrated uncertainty, an open research problem for LLMs. The prompt design, which instructs the model to answer ‘maybe’ when evidence is insufficient, may inadvertently encourage over-prediction of this class when retrieval is noisy.

Faithfulness measurement limitations. The NLI-based approach has a few constraints, starting with the 512-token limit truncates long premises, potentially cutting relevant evidence. The model was also not trained specifically for RAG faithfulness, and passage-level entailment is coarser than claim-level verification (as used in MedRAGChecker, 2026 [8]). The relatively high ‘correct but unfaithful’ rate may partly reflect limitations of the NLI model, not true unfaithfulness. Despite these constraints, the approach successfully identifies meaningful patterns.

Comparison with literature. The obtained 68.4% accuracy is comparable to the original PubMedQA best result of 68.1% (BioBERT with multi-phase fine-tuning), achieved without any fine-tuning. The retrieval findings align with MedRAG in that hybrid retrieval outperforms individual methods. BM25 remains competitive, and domain-specific retrievers like MedCPT are more effective than general-purpose ones.

6 Challenges and Limitations

API rate limiting and overload errors required robust retry logic with exponential backoff. The MedCPT encoding of 62k passages took 33 minutes on a T4 GPU, necessitating persistent storage to Google Drive for session recovery. The faithfulness evaluation required careful selection of a domain-appropriate NLI model.

Key limitations include:

1. The NLI-based faithfulness metric serves as a proxy for verification at the claim level, offering a more detailed assessment.
2. Using a single LLM (Claude Sonnet) prevents isolating generator-specific effects.
3. The 1k evaluation set limits statistical power for per-class analysis, particularly for the ‘maybe’ class (n=55).
4. The prompt was not extensively optimised through ablation, and different prompt designs might yield different accuracy-faithfulness trade-offs.

7 Conclusion and Future Work

This project deployed an end-to-end RAG pipeline for biomedical question answering with a three-dimensional evaluation framework: retrieval quality, answer accuracy, and faithfulness. The key finding is that accuracy alone is an insufficient measure of reliability for medical QA because only 45.6% of answers were both correct and faithful, with 37% of correct answers relying on ungrounded reasoning. This has direct implications for the responsible deployment of AI systems in clinical settings.

Table 9: End-to-end pipeline performance summary (test set, n=500).

Metric	BM25	MedCPT	Hybrid
Recall@5	0.846	0.884	0.924
MRR	0.755	0.754	0.834
Macro F1	0.589	0.558	0.577
Accuracy	0.672	0.652	0.684
Faithfulness	56.6%	60.8%	63.2%
Correct & Faithful	39.8%	40.4%	45.6%

Future work could improve the system in multiple areas of improvement:

- Fine-tuning the generator on biomedical QA to improve 'maybe' calibration.
- Claim-level faithfulness verification rather than passage-level NLI.
- Retrieval depth ablation (varying k) to optimise the noise-coverage trade-off.
- Multi-LLM comparison to isolate generator effects.
- Extension to multilingual medical QA for cross-jurisdictional clinical settings.

AI Usage Disclosure

- **Grammarly** was used to review and refine the text for clarity and coherence.

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